

Gabriel Barrett

Computational Biologist and Software Developer at UCONN

 Fairfield, CT



gabriel-a-barrett.github.io

Work Experience

Plant Computational Genomics | Research Specialist

January 2024 – Present | Storrs, CT

- Designed ETL pipelines for genomic, transcriptomic, phenotypic, and environmental data integration
- Developed modular tools within the command line interface for curators to parse and validate NGS data
- Integrated workflows into a centralized analysis panel, enabling users to launch parameterized jobs on a HPC system
- Automated data integration to enable meta/mega analysis

Eco-Evolution Lab | Bioinformatic Consultant

April 2024 – December 2024 | Storrs, CT

- Performed analysis on a large-scale ddRAD-seq dataset (> 1.5TB, ~1500 samples) to investigate predator-prey dynamics across spotted salamander populations in a controlled field experiment
- Conducted population genetics statistical analyses to interpret patterns of genetic variation across populations of Arctic grayling.
- Wrote a reproducible pipeline on nf-core GitHub for variant calling based on RADseq libraries
- Advised on downstream data use for publication and grant reporting

Marine Evolutionary Ecology | Research Assistant

August 2021 – December 2023 | Storrs, CT

- Executed molecular protocols for DNA extraction and library preparation, contributing to multiple genomics projects
- Performed data analysis for collaborators interested in understanding sex determination mechanisms in the purple sea urchin and advised on data use for publication
- Engineered robust scalable pipelines for variant-calling using traditional and graph-based methods

Publications

Castellano, K. R., Neitzey, M. L., Starovoitov, A., **Barrett, G. A.**, Reid, N. M., Vuruputoor, V. S., ... & O'Neill, R. J. (2025). Genome Assembly of a Living Fossil, the Atlantic Horseshoe Crab *Limulus polyphemus*, Reveals Lineage-Specific Whole-Genome Duplications, Transposable Element-Based Centromeres, and a ZW Sex Chromosome System. *Molecular Biology and Evolution*, 42(2), msaf021.

Barrett, G. A. (2024). *Variant Graphs Improve Accuracy of Downstream Analysis in Restriction Site-Associated Sequencing*. University of Rhode Island.

Education

University of Rhode Island

Master in Evolutionary Biology

May 2024 | Kingston, RI

GPA: 3.7

Thesis: Variant Graphs Improve Accuracy of Downstream Analysis in Restriction Site-Associated Sequencing

University of Connecticut

B.A. Environmental Science With

Ecology Minor

January 2021 | Kingston, RI

Skills

Programming

Proficient:

Java • R • NextFlow • Bash

Familiar:

JavaScript • HTML • CSS • SQL

Libraries/Frameworks

Git • TRIPAL • Spring Boot • R • Shiny • nf-core • Node.js

Software

ArcGIS • Excel • SLURM • NCBI datasets

Interests

Workflows Machine Learning
NGS. Visualizations Surfing
Sailing History

